BIG MODULAR 12BIT — S2 schematic: modules/8bit on an STM32G071C8T6 (two 12-bit DAC channels), 6HP. Circuit intent: docs/research.md

Signal nets are wired; supplies use power symbols.

CV IN: R4/R5/R6 level shift (−5.9..+6.0 V -> 0..3.3 V, Zin 138k), D5 clamp, C5 -> U2 MCP6001 follower -> PA0.

KNOBS P1/P2/P3: 50k linear +3V3 dividers -> PA1 / PA6 / PA7 (knob 3 is no longer the low-pass).

MCU U3 STM32G071C8T6 at 3.3 V, HSI16, no crystal.
VREF+ = +3V3 through FB1 (ADC/DAC reference).
J6 SWD: 1 +3V3 sense, 2 SWDIO, 3 SWCLK, 4 GND, 5 NRST.

OUT 1 / OUT 2: DAC1_OUT1 (PA4) / DAC1_OUT2 (PA5) -> 1k/4.7n (34 kHz step smoothing) -> TL074 x3.4 − 2.4 x VOFF
(buffered DAC 0.2..3.1 V -> +/−4.9 V). 1k 0.5 W to the jack. DC coupled: audio, CV or LFO.

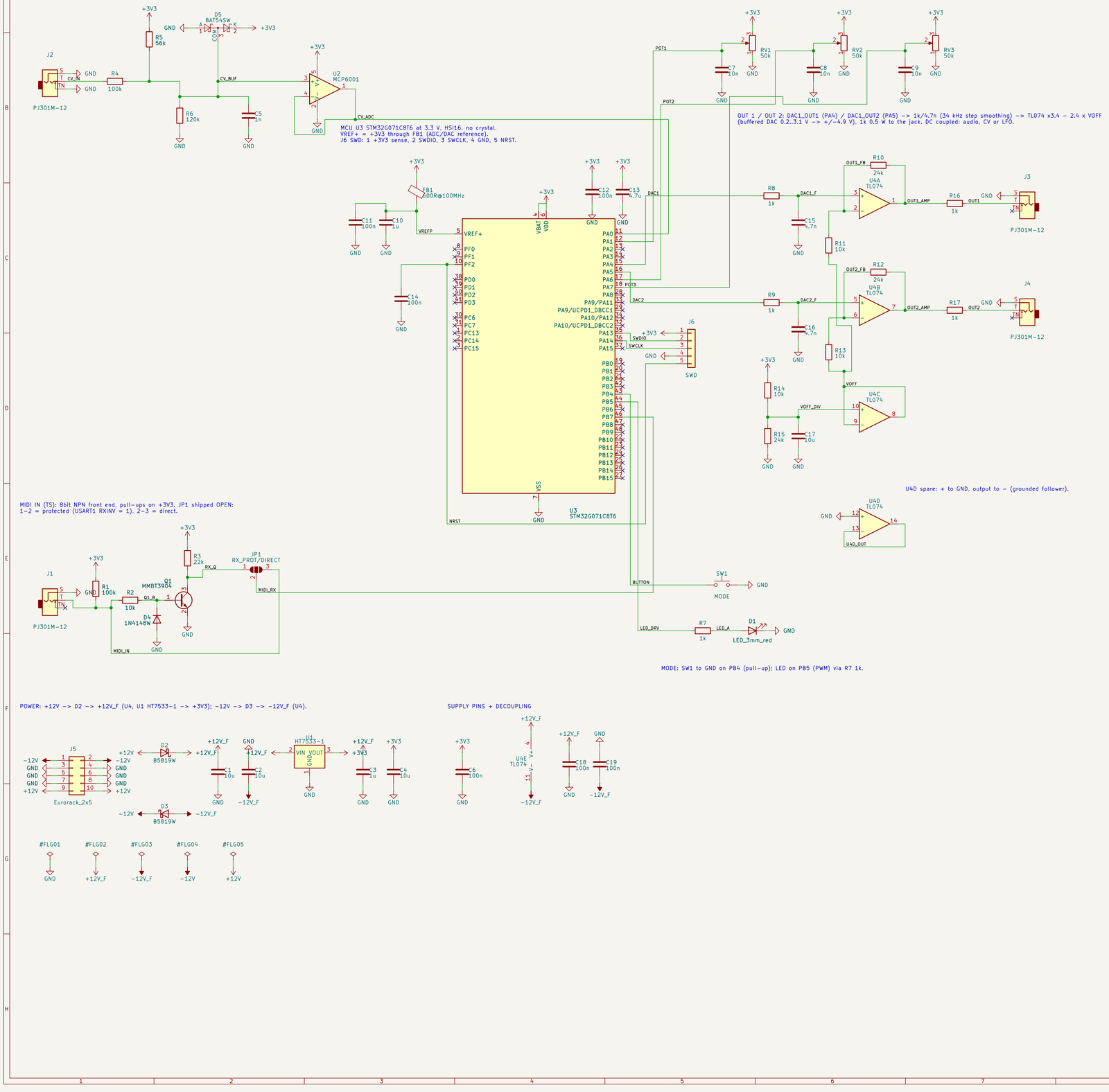
U4D spare: + to GND, output to − (grounded follower).

MIDI IN (TS): 8bit NPN front end, pull-ups on +3V3. JP1 shipped OPEN:
1-2 = protected (USART1 RXINV = 1). 2-3 = direct.

MODE: SW1 to GND on PB4 (pull-up); LED on PB5 (PWM) via R7 1k.

POWER: +12V -> D2 -> +12V_F (U4, U1 HT7533-1 -> +3V3); −12V -> D3 -> −12V_F (U4).

SUPPLY PINS + DECOUPLING



Circuit intent: docs/research.md; placement: docs/geometry.json
SCHEMATIC S2 — modules/8bit on an STM32G071C8T6: MIDI + CV in, 3 knobs, MODE, 2 x 12-bit DAC

Big Modular

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Title: Big Modular — 12bit

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